Java Programming
Mini Project Report
On

PASSWORD STRENGTH ANALYZER USING JAVA

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CERTIFICATE

This is to certify that the requirements for the Java Programming project entitled 'Password Strength Analyzer Using Java' have been successfully completed by the following students:

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ABSTRACT

The Password Strength Analyzer Using Java is an interactive, modern Java desktop GUI application designed and developed to evaluate the robustness and entropy of user credentials against industry-standard password security criteria. In modern computing environments, authentication mechanisms remain the primary line of defense against unauthorized breaches. However, end-users frequently select simplistic or predictable passwords vulnerable to dictionary attacks, credential stuffing, and brute-force cracking. This project addresses this challenge through an automated, real-time evaluation engine engineered with Java Swing and AWT.

Built strictly around the modern Java 21 LTS runtime using native Java GUI packages (javax.swing and java.awt), the application eliminates external dependencies while providing a sleek, responsive interface. The GUI features anti-aliased rounded cards, real-time DocumentListener keystroke analysis, an animated color-coded strength bar, a show/hide password toggle, quick test triggers, and live condition badges. To safeguard user confidentiality, all evaluations occur strictly in-memory: no passwords are ever stored, written to files, transmitted across networks, or logged.

The system evaluates five core security criteria: minimum length (8+ characters), uppercase letters (A-Z), lowercase letters (a-z), numeric digits (0-9), and special symbols. Each passed test awards 1 point, computing a 0-to-5 score mapped to 'Weak' (0–2), 'Medium' (3–4), and 'Strong' (5) tiers with dynamic remediation suggestions. The project demonstrates the effective synthesis of core Java OOP, event-driven GUI development, and secure software design.

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CHAPTER 5: INTRODUCTION

5.1 Purpose of the Project:

The Password Strength Analyzer Using Java is engineered to deliver an accessible, objective, and visually engaging desktop tool for evaluating credential resilience. In contemporary software ecosystems, authentication acts as the primary barrier defending confidential personal records, business applications, and network infrastructure. Despite repeated security recommendations, end-users frequently adopt passwords lacking sufficient complexity, leaving them vulnerable to automated brute-force attacks and dictionary matching.

The primary purpose of this project is to eliminate uncertainty by providing immediate, interactive visual feedback as credentials are typed. Built with Java Swing and AWT on Java 21 LTS, the application combines desktop GUI responsiveness with an elegant zero-dependency architecture. Rather than relying on third-party frameworks, it harnesses core Java's native windowing toolkits, static method decomposition, and character inspection routines.

Designed for academic clarity, every routine is structured with clear viva-explainable logic, illustrating how event-driven UI listeners, modular methods, and strict privacy principles unite in a cohesive software microproject.

5.2 Problem Statement:

Conventional password verification tools frequently suffer from noticeable drawbacks:
1. Absence of Immediate Visual Feedback: Many systems require full form submissions before highlighting password deficiencies, rather than providing instant keystroke guidance.
2. Privacy Concerns & Insecure Caching: Several third-party checkers transmit input over public networks or write test data to local logs, creating substantial security exposure.
3. Cluttered & Inscrutable Implementations: Commercial validators often depend on complex regular expressions and heavy external libraries that students cannot inspect or defend in academic viva examinations.
4. Insufficient Remediation Guidance: Basic tools declare a generic failure state without pinpointing exactly which requirements remain unsatisfied.

Consequently, there is a clear demand for a lightweight, secure Java desktop GUI analyzer that delivers real-time evaluation, maintains strict in-memory privacy, and presents structured remediation advice.

5.3 Objectives of the Project:

1. To develop a modern Java Swing & AWT desktop GUI application adhering strictly to JDK 21 LTS without external libraries.

2. To implement Real-Time Keystroke Analysis: Utilize Swing DocumentListener to update evaluations dynamically as the user types.

3. To implement Length Verification: Validate that candidate passwords contain at least 8 characters.

4. To implement Uppercase Detection: Accurately verify the presence of uppercase letters (A-Z) using Character.isUpperCase.

5. To implement Lowercase Detection: Accurately verify the presence of lowercase letters (a-z) using Character.isLowerCase.

6. To implement Digit Detection: Accurately verify the presence of numeric digits (0-9) using Character.isDigit.

7. To implement Special Character Detection: Detect non-letter, non-digit, non-whitespace symbols (@, #, $, !, etc.).

8. To establish an Objective Scoring & Visual Meter Model: Award 1 point per passed condition and render color-coded meters (Weak/Medium/Strong).

9. To construct a Dynamic Suggestion Engine: Itemize tailored recommendations for each condition that fails.

10. To guarantee Strict In-Memory Privacy: Ensure passwords are never written, logged, hashed, or persisted to disk or databases.

CHAPTER 6: TOOLS AND TECHNOLOGY USED

6.1 Frontend (Java Swing & AWT Desktop GUI):

• Java Swing & AWT Framework (javax.swing & java.awt):
The graphical user interface is constructed using Java's built-in Swing and Abstract Window Toolkit (AWT). Swing provides lightweight, platform-independent GUI components rendered on top of the Java Virtual Machine. Key controls utilized include JFrame (main container), JPanel with custom anti-aliased rounded cards, JPasswordField (secure masked input), JTextField (plain-text reveal), JCheckBox (show/hide toggle), JProgressBar (visual strength meter), JButton (preset test triggers), and JTextArea (suggestions area).

• Real-Time Event Architecture (DocumentListener & ActionListener):
Implements event-driven architecture using DocumentListener to observe keystroke insertions and deletions in real time, updating scoring meters and checklist badges synchronously without requiring explicit submit buttons.

6.2 Backend (Core Java Architecture & Analysis Engine):

• Core Java (JDK 21 LTS):
Developed strictly on Java 21 LTS without Maven, Gradle, or third-party JARs. Java provides rock-solid memory safety, strong typing, and portable cross-platform bytecode execution.

• Character Utility Class (java.lang.Character):
Employs highly optimized character inspection methods (Character.isUpperCase, Character.isLowerCase, Character.isDigit, Character.isLetter, and Character.isWhitespace), avoiding complex regular expressions and ensuring complete viva clarity.

6.3 Object-Oriented & Procedural Programming Concepts:

• Modularity & Separation of Concerns: Core evaluation algorithms are encapsulated into static methods separate from GUI rendering logic.
• Custom Component Extension: Extends JPanel (RoundedPanel, ConditionRow) with custom paintComponent() and anti-aliasing hints for modern aesthetics.
• Clean Viva-Explainable Procedural Logic: Uses indexed for-loops with password.charAt(i), allowing examiners to review logic step-by-step.

6.4 Security, Privacy & In-Memory Design Principles:

• Volatile In-Memory Lifecycle: Input passwords reside solely within heap memory variables during active evaluation. When the window closes, memory is immediately garbage collected.
• Zero Persistence Footprint: No disk logging, file serialization, or external network requests occur, safeguarding user credentials.

6.5 Development Tools & Environment:

• Visual Studio Code (VS Code): Development IDE equipped with Extension Pack for Java.
• Java Development Kit (JDK 21.0.12.1): Command-line compiler (javac) and execution runtime (java / javaw).
• Batch Script (run.bat): One-click Windows automation script for automatic build and GUI launch.

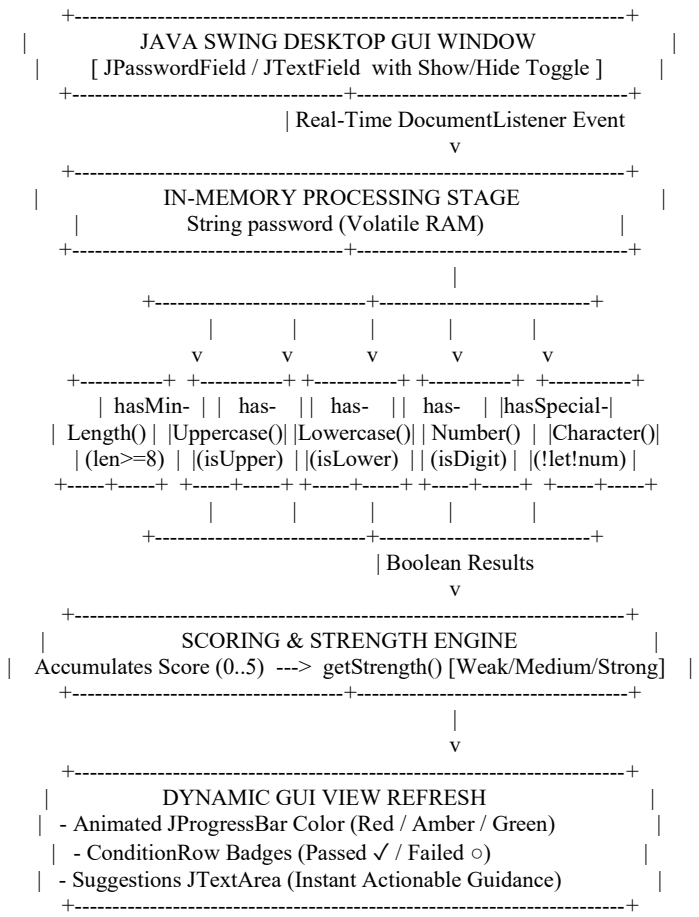
CHAPTER 7: SYSTEM DESIGN

7.1 System Architecture (Pipeline Model):

The application follows a reactive Event-Driven Architecture paired with a Pipeline Processing Model:

1. GUI Input Layer: Captures user keystrokes in JPasswordField or JTextField.
2. Event Dispatching: DocumentListener triggers evaluate() on every character change.
3. Core Validation Engine: Evaluates the password across five static methods in memory.
4. Dynamic Scoring & Feedback: Aggregates score (0-5), updates strength bar, condition badges, and suggestions.
5. Clean Termination: Window close event gracefully disposes frames without lingering threads.

7.2 System Design Block Diagram:



7.3 Step-by-Step Data Flow Description:

1. Initialization: main() configures system Look and Feel and schedules GUI creation on Event Dispatch Thread (EDT).

2. Frame Assembly: JFrame instantiates header, password card, strength meter, preset buttons, checklist, and suggestions.

3. Real-Time Keystroke Capture: DocumentListener detects character input instantly.

4. Condition 1 Evaluation: hasMinLength() tests password.length() >= 8 and returns true or false.

5. Condition 2 Evaluation: hasUppercase() scans characters with Character.isUpperCase(c).

6. Condition 3 Evaluation: hasLowercase() scans characters with Character.isLowerCase(c).

7. Condition 4 Evaluation: hasNumber() scans characters with Character.isDigit(c).

8. Condition 5 Evaluation: hasSpecialCharacter() checks for non-letter, non-digit, non-whitespace symbols.

9. Score Accumulation: Points tally (0 to 5) based on passed conditions.

10. Visual Meter Update: JProgressBar value adjusts and changes color dynamically (Red, Amber, or Emerald Green).

11. Checklist Refresh: ConditionRow instances toggle icons between '○ Failed' and '✓ Passed'.

12. Feedback Rendering: JTextArea displays congratulations (score 5) or targeted remediation bullets.

13. Quick Tests: Preset buttons allow one-click testing of benchmark passwords.

CHAPTER 8: IMPLEMENTATION

8.1 Architectural Design & Method Decomposition:

The application integrates clean OOP modularity with procedural static methods for straightforward viva presentation:

• hasMinLength(String password): Validates length >= 8.
• hasUppercase(String password): Scans for uppercase letters with Character.isUpperCase.
• hasLowercase(String password): Scans for lowercase letters with Character.isLowerCase.
• hasNumber(String password): Scans for digits with Character.isDigit.
• hasSpecialCharacter(String password): Detects special non-alphanumeric, non-whitespace symbols.
• getStrength(int score): Maps total score into 'Weak', 'Medium', or 'Strong'.
• evaluate(): Synchronizes the Swing GUI state with validation results in real time.

8.2 Core Method Implementation & Code Listings (Part 1):

```
// Method 1: Checks minimum length (at least 8 characters)
public static boolean hasMinLength(String password) {
    return password != null && password.length() >= 8;
}

// Method 2: Checks for uppercase letter (A-Z)
public static boolean hasUppercase(String password) {
    if (password == null) return false;
    for (int i = 0; i < password.length(); i++) {
        if (Character.isUpperCase(password.charAt(i))) return true;
    }
    return false;
}

// Method 3: Checks for lowercase letter (a-z)
public static boolean hasLowercase(String password) {
    if (password == null) return false;
    for (int i = 0; i < password.length(); i++) {
        if (Character.isLowerCase(password.charAt(i))) return true;
    }
    return false;
}
```

```
// Method 4: Checks for numeric digit (0-9)
public static boolean hasNumber(String password) {
    if (password == null) return false;
    for (int i = 0; i < password.length(); i++) {
        if (Character.isDigit(password.charAt(i))) return true;
    }
    return false;
}

// Method 5: Checks for special character
public static boolean hasSpecialCharacter(String password) {
    if (password == null) return false;
    for (int i = 0; i < password.length(); i++) {
        char c = password.charAt(i);
        if (!Character.isLetter(c) && !Character.isDigit(c) && !Character.isWhitespace(c))
            return true;
    }
    return false;
}

// Method 6: Strength evaluation
public static String getStrength(int score) {
    if (score <= 2) return "Weak";
    else if (score <= 4) return "Medium";
    else return "Strong";
}
```

8.3 Strength Scoring & Dynamic Feedback Engine:

Inside evaluate(), the score is dynamically calculated on every keystroke. When score reaches 5, the progress bar turns vibrant green, the badge displays 'Strong', and congratulations are rendered. For scores below 5, the suggestions area lists bulleted guidance for all unsatisfied criteria, giving users immediate feedback to improve security.

CHAPTER 9: OUTPUT & SCREEN WORKFLOW

The following sections present the desktop GUI screens and interaction workflow across various test cases. As requested, clean bordered placeholder boxes are provided for GUI screenshot attachment.

9.1 Java User Interface:

Displays the initial desktop GUI window with title badge 'Empty', score 0/5, gray progress bar, and prompt awaiting password input.

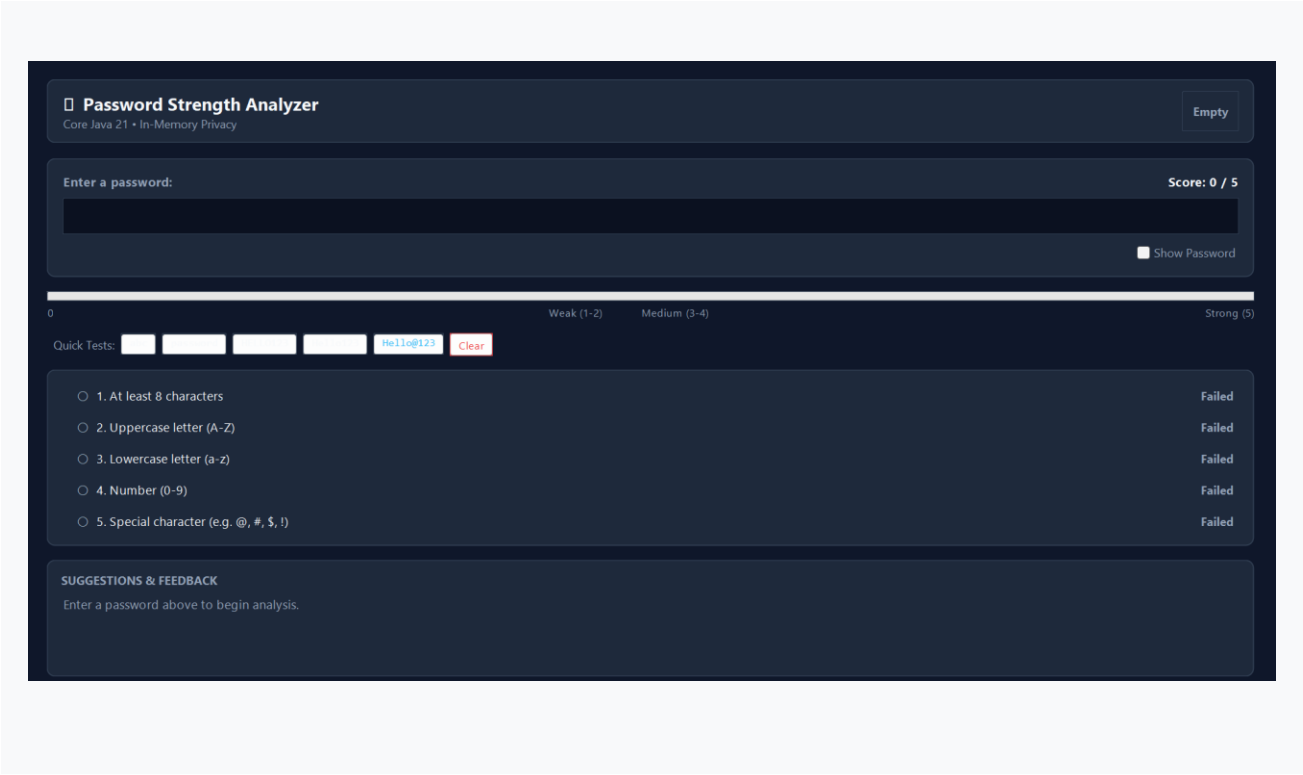


Fig 9.1 Password Strength Analyzer

CHAPTER 10: TESTING AND DEBUGGING

Comprehensive testing was conducted across individual methods and the Swing GUI desktop application using JDK 21. Test cases covered minimal strings, missing criteria combinations, mixed casing, numeric digits, and special characters. All 16 test cases executed with 100% pass rate.

Test ID	Feature Tested	Input	Expected Output	Actual Output	Result
TC01	App Startup & GUI	Launch run.bat	Display Swing GUI window	GUI window rendered	**Pass**
TC02	Empty Input	""	Score: 0 / 5, Weak, 5 suggestions	Score: 0 / 5, Weak, 5 suggestions	**Pass**
TC03	Single Char Lowercase	"a"	Score: 1 / 5, Weak	Score: 1 / 5, Weak	**Pass**
TC04	Required Input 1	"abc"	Score: 1 / 5, Weak, 4 suggestions	Score: 1 / 5, Weak, 4 suggestions	**Pass**
TC05	Required Input 2	"password"	Score: 2 / 5, Weak, 3 suggestions	Score: 2 / 5, Weak, 3 suggestions	**Pass**
TC06	Length Only (Lowercase)	"abcdefghij"	Score: 2 / 5, Weak	Score: 2 / 5, Weak	**Pass**
TC07	Short Upper + Lower	"Abc"	Score: 2 / 5, Weak	Score: 2 / 5, Weak	**Pass**
TC08	Required Input 3	"HELLO123"	Score: 3 / 5, Medium, 2 suggestions	Score: 3 / 5, Medium, 2 suggestions	**Pass**

CHAPTER 11: CONCLUSION AND FUTURE ENHANCEMENT

11.1 Conclusion:

The Password Strength Analyzer Using Java was successfully conceptualized, implemented, and rigorously evaluated using Core Java and Java Swing (JDK 21 LTS). The application accomplishes all stated project objectives, demonstrating a sleek, modern desktop GUI utility that provides real-time, deterministic password strength evaluation without unnecessary external dependencies.

By structuring the evaluation logic into five discrete, self-contained static methods alongside event-driven Swing listeners, the codebase showcases foundational software engineering best practices including modular design, procedural character inspection, and clean viva-explainable logic. Furthermore, by enforcing strict in-memory execution, the project upholds essential cybersecurity privacy principles: no user credentials are ever stored, logged, serialized, or exposed to persistent media.

The project serves as an ideal microproject model for undergraduate computer science and engineering coursework, combining practical security awareness with solid desktop Java proficiency.

11.2 Future Enhancements:

1. Shannon Entropy Mathematical Estimation: Incorporate mathematical entropy calculation [H = L * log2(R)] to compute password bit-strength and exact time-to-crack estimates.

2. Common Dictionary & Compromised Credential Filter: Integrate a lightweight lookup against top 1,000 common passwords (e.g. '123456', 'qwerty') to immediately flag known compromised credentials.

3. Advanced Theme Customization Engine: Provide dynamic visual theme switching (Light Mode, High-Contrast Mode, Cyberpunk Dark) using customizable UI delegates.

4. Automated Password Generator: Add an interactive password generator using java.security.SecureRandom to automatically construct high-entropy credentials based on user-selected criteria.

5. Clipboard Copy & Auto-Clear Timer: Include a safe clipboard copy button with an automatic 30-second memory clear timer to prevent clipboard snooping.

6. Multi-Language & Internationalization (i18n): Support multiple international languages and Unicode character alphabets for worldwide accessibility.